Day 1 — ML Foundations and Mathematical Language

Objective

Build the vocabulary and mathematical intuition needed to understand why ML algorithms work.

Topics and detailed notes

- Machine learning as parameter estimation and generalization rather than pattern memorization.
- Supervised learning: inputs X and target y. Regression predicts a continuous quantity; classification predicts a class or probability.
- Unsupervised learning: structure is inferred without a supplied target.
- Features, target, observation, parameter, hyperparameter, estimator, loss, objective, inference, artifact.
- Training data is used to learn parameters. Validation data supports decisions about model configuration. The final test set estimates performance after decisions are frozen.
- Generalization: performance on unseen data from the intended deployment distribution.
- Baseline: the simplest meaningful reference against which a model must prove value.
- Batch inference vs online inference.
- Offline training vs online prediction.

Mathematics / formulas to know

Vector: an ordered set of values representing an observation or parameters.

Matrix multiplication combines observations and coefficients; Xw produces one score per observation.

Dot product: a weighted sum of feature values.

Derivative: local rate of change. Gradient: vector of partial derivatives.

Gradient descent update: $\theta \leftarrow \theta - \alpha \nabla J(\theta)$.

Learning rate α controls the size of the update.

1. Mean

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

2. Weighted Mean

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

Useful when observations have different importance/weights.

3. Variance

Population variance:

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

Sample variance:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

4. Standard Deviation

$$\sigma = \sqrt{\sigma^2}$$

It measures the typical spread of values around the mean.

5. Z-Score

$$z = \frac{x - \mu}{\sigma}$$

This tells you how many standard deviations an observation is away from the mean.

6. Covariance

$$Cov(X, Y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

It describes how two variables vary together.

7. Pearson Correlation

$$r = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

with:

$$-1 \leq r \leq 1$$

8. Min-Max Scaling

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Usually maps values into:

$$[0, 1]$$

9. Standardization

$$x' = \frac{x - \mu}{\sigma}$$

After standardization, the feature approximately has:

$$\mu = 0, \qquad \sigma = 1$$

10. Euclidean Distance

For two points:

$$x = (x_1, x_2, \ldots, x_n)$$

and

$$y = (y_1, y_2, \ldots, y_n)$$

the distance is:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

You'll need this later for algorithms such as **KNN and clustering.**

11. Basic Probability

$$P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

And conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

12. Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

This becomes important later for **Naive Bayes.**

Implementation pattern

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Problem → mathematical formulation → data → objective/loss → optimization → model →
evaluation → decision.
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Experiments to perform

- Create a tiny synthetic regression dataset and manually compute a prediction from a weight vector.
- Calculate a dot product by hand and with NumPy.
- Implement a simple loss calculation.
- Implement one gradient-descent update manually before using a library implementation.

End-of-day deliverables

- Change the learning rate and observe convergence.
- Use poorly scaled features and observe optimization behavior.
- Compare a constant baseline with a learned model.

Mastery questions

1. One notebook explaining the ML vocabulary.
2. A small NumPy gradient-descent experiment.
3. A written explanation of training vs validation vs test.